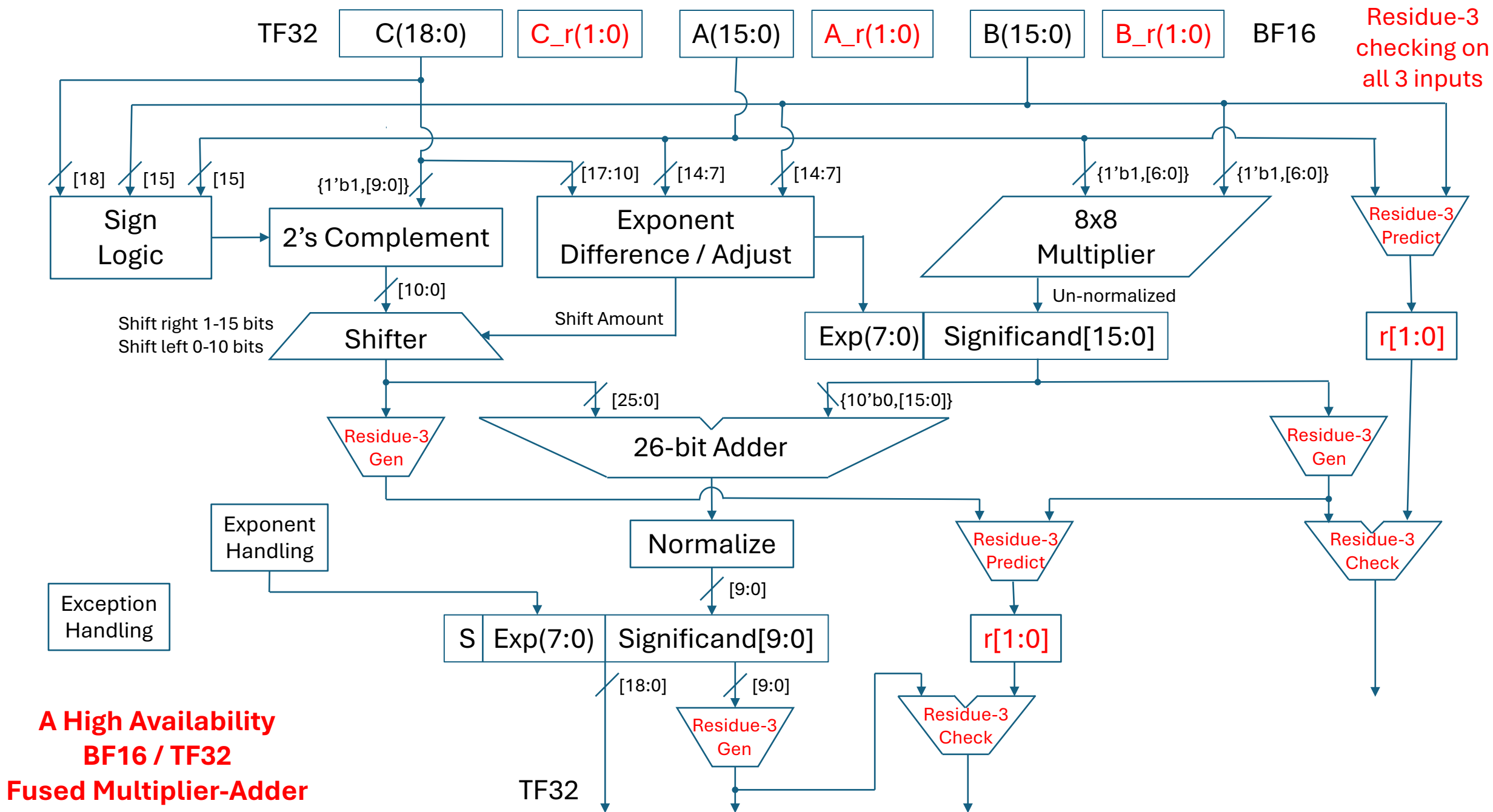


A High Availability BF16 / TF32 Fused Multiplier-Adder

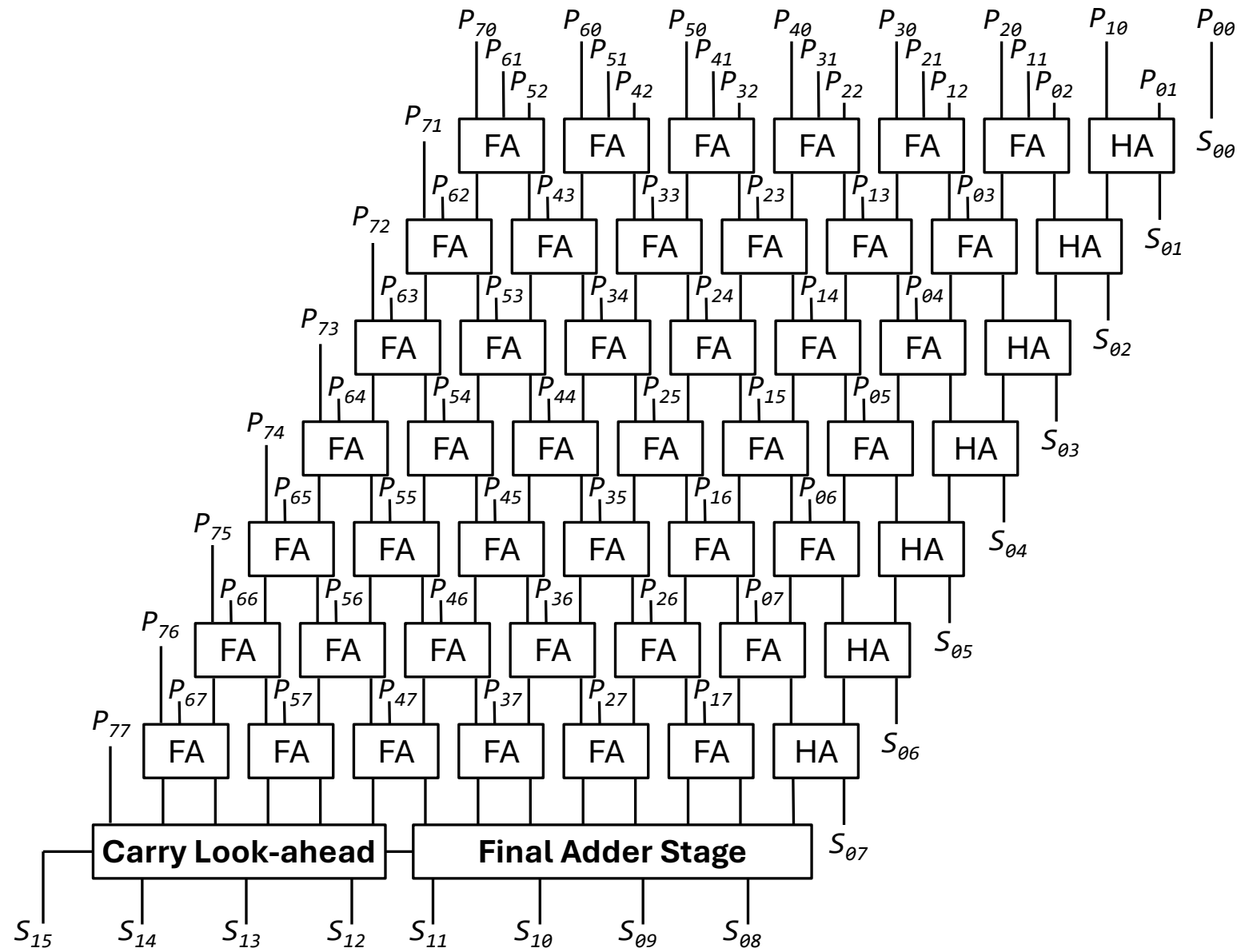
Donald McCauley TechAnalytics LLC January 5, 2026



Unsigned 8-bit Multiply

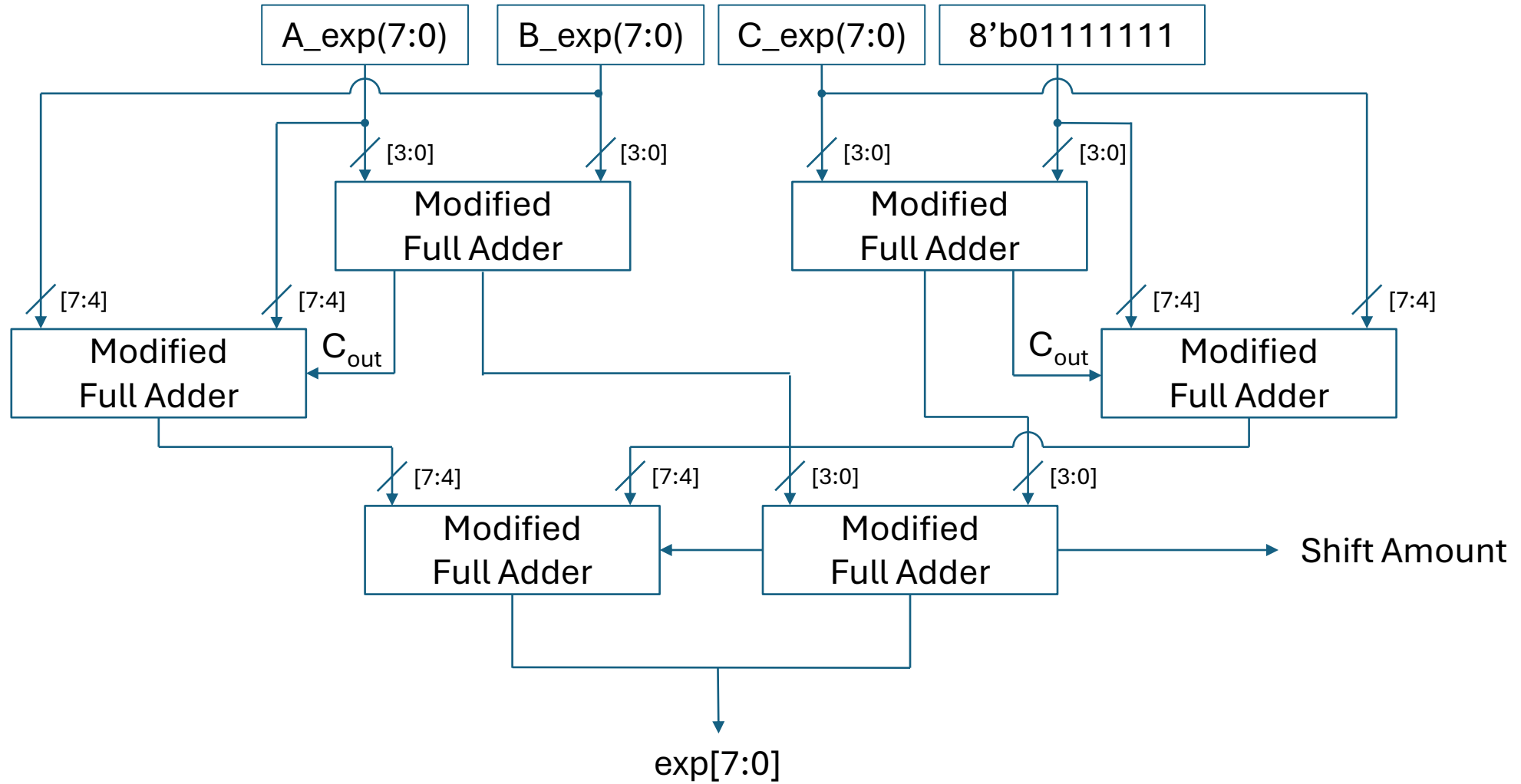
								A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
								x	B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀
									P ₇₀	P ₆₀	P ₅₀	P ₄₀	P ₃₀	P ₂₀	P ₁₀	P ₀₀
							P ₇₁	P ₆₁	P ₅₁	P ₄₁	P ₃₁	P ₂₁	P ₁₁	P ₀₁		
						P ₇₂	P ₆₂	P ₅₂	P ₄₂	P ₃₂	P ₂₂	P ₁₂	P ₀₂			
					P ₇₃	P ₆₃	P ₅₃	P ₄₃	P ₃₃	P ₂₃	P ₁₃	P ₀₃				
				P ₇₄	P ₆₄	P ₅₄	P ₄₄	P ₃₄	P ₂₄	P ₁₄	P ₀₄					
			P ₇₅	P ₆₅	P ₅₅	P ₄₅	P ₃₅	P ₂₅	P ₁₅	P ₀₅						
		P ₇₆	P ₆₆	P ₅₆	P ₄₆	P ₃₆	P ₂₆	P ₁₆	P ₀₆							
	P ₇₇	P ₆₇	P ₅₇	P ₄₇	P ₃₇	P ₂₇	P ₁₇	P ₀₇								
S ₁₅	S ₁₄	S ₁₃	S ₁₂	S ₁₁	S ₁₀	S ₀₉	S ₀₈	S ₀₇	S ₀₆	S ₀₅	S ₀₄	S ₀₃	S ₀₂	S ₀₁	S ₀₀	

Source: “Design and implementation of an eight bit multiplier using Twin Precision technique and Baugh-Wooley algorithm” [9]



Source: [10] “Design of Delay-Efficient Carry-Save Multiplier by Structural Decomposition of Conventional Carry-Save Multiplier...”

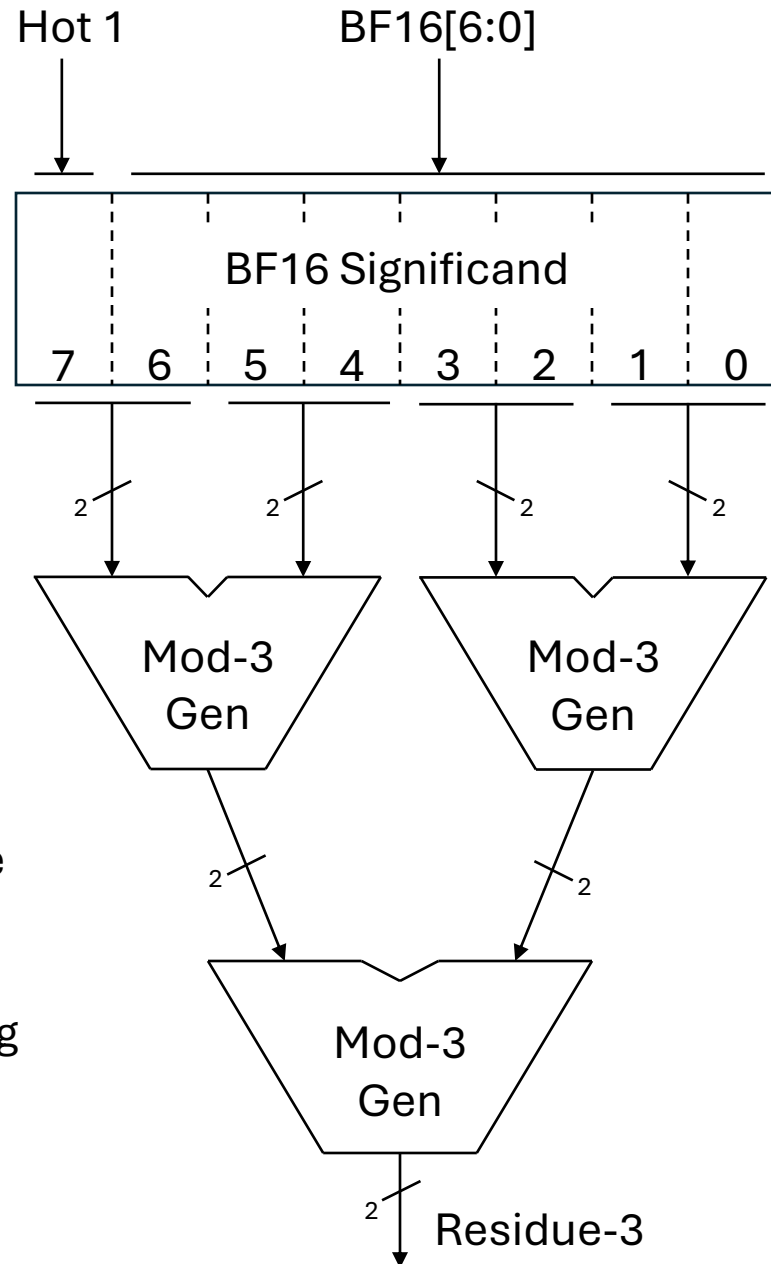
Exponent Difference / Adjust Logic



Source: 'A Bfloat16 Fused Multiplier-Adder' [7]

Residue-3 checking detects 100% of single bit errors and 66% of multiple bit errors.

The predicted residue of the product of two BF16 significands is simply the product of the corresponding residues.



BF16 Residue-3 Generation

Estimated cost in gates & delay:

~12 gates per Mod-3 generator;
~4 logic levels per Mod-3 generator.

A total of ~13 MOD-3 generators are required to generate residues for both 8-bit inputs as well as their 16-bit product. (Logic can be combined with input Residue-3 checking.)

Total cost of residue checking: TBD

Modulo-3 Generator

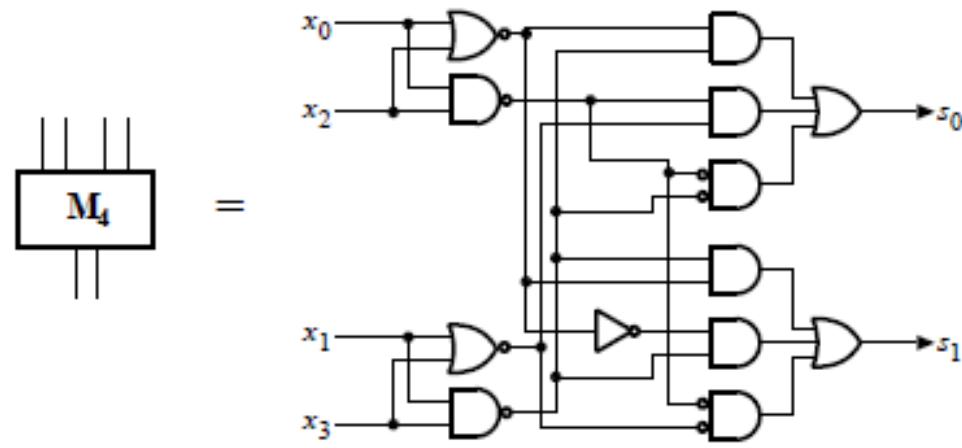


Figure 2: Cell M_4 of the 4-input generator mod 3.

Source: "VLSI IMPLEMENTATION AND COMPLEXITY COMPARISON OF RESIDUE GENERATORS MODULO 3" [8]

Backup

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